

Nikkei 225

Nikkei 225

Valuation Report

March-24

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1.1 Background

- The Nikkei 225 is a major stock market index comprising 225 prominent companies on the Tokyo Stock Exchange, selected based on stock price rather than market capitalization.
- The Nikkei 225 began to be calculated on 7 September 1950.
- The sheer size of the Japanese economy makes the Nikkei 225 a key measure of stock market activity in Asia
- The Nikkei 225, or Nikkei Stock Average, includes companies from 36 different industries.
- These industries Include: Technology, Consumer goods, Financials, Materials, Capital goods, Transportation, Utilities, Pharmaceuticals, Electric machinery, Automobiles and auto parts, Precision instruments, and Communications.

1.2 Key Stats

- Japan's Nikkei 225 stock index closed above 40,369 on March 28, setting another record high after climbing last month.
- The broader Tokyo Stock Price Index, or TOPIX, has also rallied this year, approaching all-time highs. The Nikkei gained 28% and the TOPIX 25% last year.
- The Average historical CAGR returns provided by the Index for various periods are:

a) Nikkei CAGR Return 70 Years- 6.72%	e) Nikkei CAGR Return 10 Years- 7.82%
b) Nikkei CAGR Return 50 Years- 4.55%	f) Nikkei CAGR Return 5 Years- 8.04%
c) Nikkei CAGR Return 30 Years- 2.39%	g) Nikkei CAGR Return 3 Years- 15.68%
d) Nikkei CAGR Return 20 Years- 4.70%	

1.3 Composition of Nikkei 225

- Nikkei 225 Index has Stocks of 225 Companies.
- Few Notable companies are- Fast Retailing Co.Ltd,Tokyo Electron Ltd,Advantest Corporation,KDDI Corporation,Shin-Etsu Chemical Ltd,Darkin Industries Ltd,FANUC Corporation,Terumo Corporation.
- The index comprises wide range of sectors such as Utilities, Pharmaceuticals, Electric machinery, Automobiles and auto parts, Precision instruments, and Communications, Consumer goods, Financials, Materials, Capital goods, Transportation.

Sector	Weightage %
Communications	9.45%
Consumer Discretionary	20.97%
Consumer Staples	5.42%
Energy	0.28%
Financials	2.66%
Health Care	10.27%

Sector	Weightage %
Industrials	18.39%
Information Technology	24.23%
Materials	6.59%
Utilities	0.17%
Real Estate	1.57%

2.1 Context

- Two important structural changes are playing out in the Japanese stock market: The country is shifting to an inflationary economy after years of deflation, and corporate governance reforms are taking root.
- Last month, the benchmark index surpassed its 1989 record of 38,915.8, set as Japan's economy was on the precipice of an asset crash that kicked off several "lost decades" of economic stagnation.
- Japan's overall economy has not matched the fortunes of its stock market, with structural challenges, including a shrinking population and rigid labour force, weighing on growth.
- The Valuation report aims to determine the valuation of the Index in the above context.
- The Index has been valued considering the dividends paid, buyback yield, Future Earnings Growth, Equity Risk Premium and India's 10 Year Government Bond Yield which serves as an appropriate proxy for Risk Free Rate. The valuation of the index is based on the methodology used by Prof. Aswath Damodaran (Dean of Valuation- NYU Stern) and his pronouncements on the Discounted Cash Flow approach.
- The Report is prepared using assumptions. Thus, the interpretation of results shall be subjective and based on the users' requirements. The report provides a broader idea on whether the Nikkei index is Overvalued, Undervalued or Fairly valued which are derived primarily from the cash flows (Dividends and Buy Backs) in the hands of investors/shareholders.
- The index's valuation date shall be "28th March 2024" and all the data used for the purpose of valuation are till 28th March 2024.

3.1 Pillars of the Valuation

- The Valuation approach used to value the Nifty 50 Index is the Discounted Cash Flow approach. The key pillars being a.) Free Cash Flow to Equity (FCFE) b.) Earnings Growth c.) Risk Free Rate and d.) Equity Risk Premium.

3.2 Free Cash Flow to Equity Holders

- Cash flows being one of the key ingredients in DCF valuation implies the extent of the equity holders right on the cash flows. Cash flows implies Free Cash Flows to Equity holders i.e., FCFE.
- However, it becomes a laborious task to derive FCFE of all the companies in the index and hence, a convenient and effective approach has been followed where the Dividend pay-outs and Buybacks are substituted for FCFE being the ultimate cash flows in the hands of shareholders.
- The rationale behind the above consideration is that whatever the cash a firm generates, it shall be ultimately paid to the shareholders in the long run either on liquidation or after the firm has attained maturity in its business verticals either through dividends or buybacks.
- The dividend yield has been sourced from Nikkei Website and the buyback data not readily available.
- In the absence of buyback data, a multi-step approach was devised. Firstly, dividend yield data from Nikkei Website was utilized. Next, earning yield was computed using EPS and index price. Then, buyback yield was derived incorporating dividend yield, earning yield, and P/E ratio. Finally, buyback amount was estimated by multiplying buyback yield with index price. This method ensures transparency and rigor in estimating the buyback amount despite unavailable data, enhancing the depth of analysis.

3.Approach and Methodology

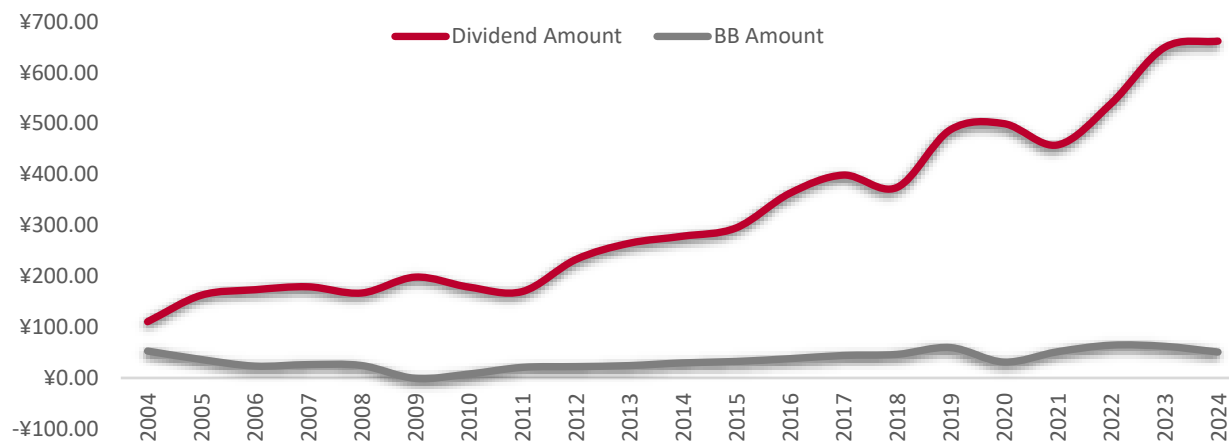
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3.2 Free Cash Flow to Equity Holders

- The below representations provides the insights on % of dividend yield and buyback yield in total yield and average Nikkei 250 yield over 20, 15, 10, 5 and 3 years time frame.

Time Frame(Years)	Dividend Yield	Buyback Yield	Total Yield
20	1.72%	0.18%	1.89%
15	1.83%	0.18%	2.01%
10	1.82%	0.19%	2.01%
5	1.81%	0.17%	1.98%
3	1.88%	0.19%	2.07%

Historical Dividend&Buyback



3.Approach and Methodology

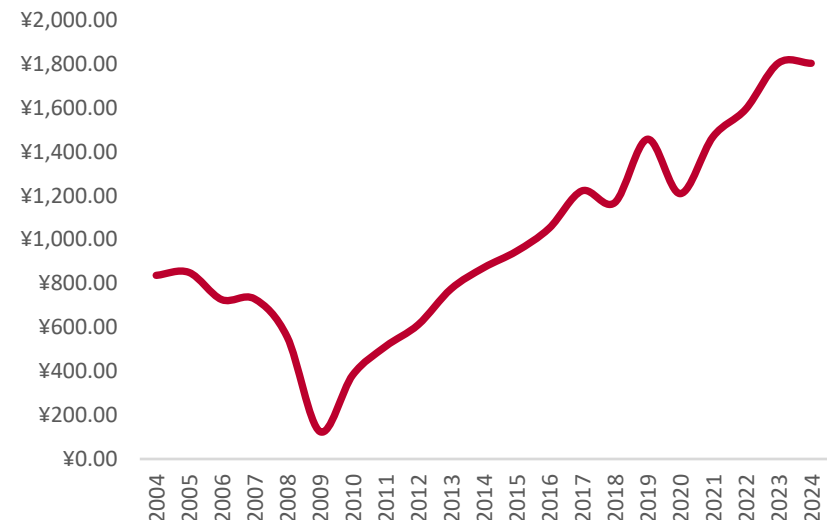
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3.3 Earning Growth

- Earnings of Nikkei 225 companies are function of Index EPS and Index's price.
- Earnings are computed using P/E ratio formula where the earnings are obtained by dividing Index's PE Ratio by the price of Nikkei i.e. Nikkei points.
- The Nikkei PE ratio and the corresponding price levels are sourced from Nikkei website and then earnings are derived.
- The data in Nikkei website provides daily data of Nifty prices and PE ratios. Yearly Average PE range and yearly average Nifty price levels have been considered for the sake of brevity and convenience.
- EPS CAGR for 3Y, 5Y, 10Y and 15Y were calculated and average of those has been considered for the final growth figures.
- For computing future cash flows, Average EPS CAGR 3Y has been considered over 5Y, 10Y and 15Y as 3Y provides the best representation of the current market trends along with factoring the slowdown caused by Covid-19, geopolitical tensions, distressed supply chains and so on.
- However, there has been no significant difference in the growth rates between the above mentioned time frames. The same has been depicted in the table beside

Average EPS CAGR	Earnings Growth%
15	7.40%
10	7.77%
5	4.71%
3	4.44%

Nikkei EPS



3.Approach and Methodology

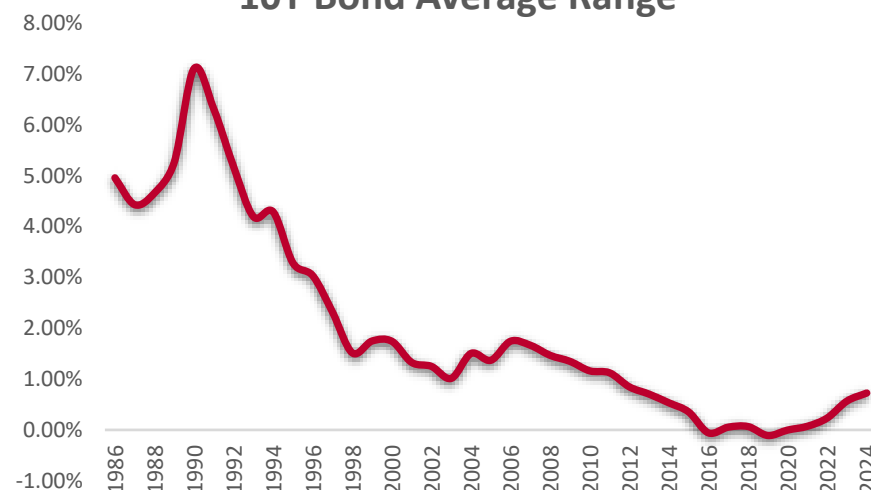
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3.4 Risk Free Rate

- Risk free rate implies the guaranteed minimum return that any Investor shall expect for Investing in a particular country.
- In valuation practice, we consider an economy's 10Y Government yield as a Risk free rate. But, the most appropriate manner shall be to deduct a country's default spread to factor in sovereign default probabilities.
- As the Historical default spread data are not available and it is a tedious task to account for the same and hence Japan's 10Y Government Bond Yield has been considered as Risk Free rate.
- Given the fact that the Japan's 10Y Government Bond Yield has varied largely from ~1.00% to ~.70% because of mature economy, for the purpose of valuation, the latest Risk free rate as on March 28, 2024 has been considered to satisfy the twin objectives of enabling our valuation to be a forward looking one and to depict the general economic scenario of Japan's economy.

Average Rate for	Japan 10Y Govt.Bond Yield
30	1.09%
20	0.69%
10	0.19%
5	0.32%
2024	0.73%

10Y Bond Average Range



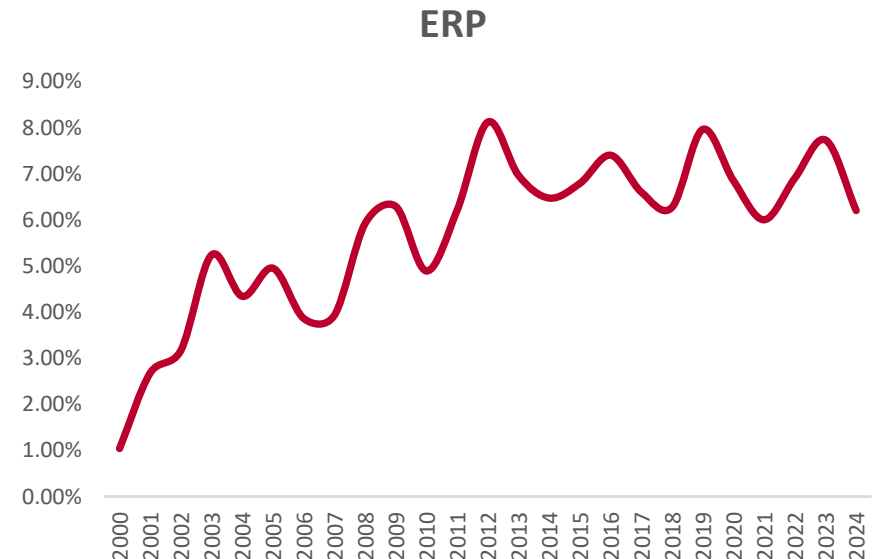
3.Approach and Methodology

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3.5 Market Risk Premium

- Market Risk Premium is the additional return that an investor expects over and above the Risk Free Rate for investing in a particular market which is usually deemed risky.
- It is a country specific risk/ market specific risk.
- When the Market Risk premium is more, it indicates cautious market stance where the investors are in a defensive mode and expects usual than higher returns to invest in a market.
- It also implies that the investors are willing to pay lower prices despite the constant cash flows.
- The data of Market Risk premium for the valuation has been sourced from Market Premia website.
- For the purpose of our Valuation, we have considered the latest Market Risk Premium to account for the latest developments in Indian Equity markets.

Average ERP for:	Equity Risk Premium
20	6.32%
15	6.76%
10	6.87%
5	6.74%
2024	6.20%



3.6 Time Frame for the Valuation Report

- The Valuation date for the purpose of the report shall be 28st March, 2024.
- All the averages for the data such as Risk Free Rate, Market Risk Premium, Nikkei 225 historical returns, EPS Growth, dividend yield have been calculated from FY 1949-50 to FY 2023-24.
- The valuation assessment has been made by comparing the derived Nikkei 225 value as per our valuation and the Closing price of Nikkei as on 28st March,2024.

3.7 Beta

- Beta while valuing index shall be considered to be '1' as we are valuing the barometer which broadly represents the market.

4. Valuation

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4.1 Original Case Valuation

Valuing The Nikkei 250

Index	Previous close	Ticker Symbol	Intrinsic value
 NIKKEI 225 INDEX	¥ 40,168.07	Nikkei 250	¥44,262.89
Key Inputs		Assumptions	Undervalued The market implied fair value of Nikkei is Rs. 44263. The Nikkei is currently trading at 40169. A 9.25% appreciation is expected from this level.
Date	30-03-2024	30-03-2024	
Current Nikkei Level	¥ 40,168.07	¥ 40,168.07	
Total Yield	5 Years	5.09%	
Expected Growth	3 Years	4.44%	
Risk-Free Rate	Letest	0.73%	
Equity Risk Premium	Letest	6.20%	
Cost of Equity		6.93%	

Year	Expected Dividend&Buyback	Cumulative PV Factor (Risk-free Rate + Equity Risk Premium)	Present Value of Expected Dividends & BuyBacks
2024	¥ 2,135.77	0.935	¥1,997.43
2025	¥ 2,230.56	0.875	¥1,950.96
2026	¥ 2,329.57	0.818	¥1,905.57
2027	¥ 2,432.96	0.765	¥1,861.24
2028	¥ 2,540.95	0.715	¥1,817.94
2029	¥ 2,653.73	0.669	¥1,775.65
2030	¥ 2,771.51	0.626	¥1,734.34
2031	¥ 2,894.52	0.585	¥1,693.99
2032	¥ 3,022.99	0.547	¥1,654.58
2032 - ∞	¥ 50,922.03	0.547	¥ 27,871.21

4. Valuation

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4.1 Fair Valuation Case Scenario

Valuing The Nikkei 250

Index	Previous close	Ticker Symbol	Intrinsic value
 NIKKEI 225 INDEX	¥ 40,168.07	Nikkei 250	¥40,525.37
Key Inputs		Assumptions	Fairly Valued The market implied fair value of Nikkei is Rs. 40526. The Nikkei is currently trading at 40169. A 0.88% adjustment is expected from this level.
Date	30-03-2024	30-03-2024	
Current Nikkei Level	¥ 40,168.07	¥ 40,168.07	
Total Yield	5 Years	5.09%	
Expected Growth	3 Years	4.44%	
Risk-Free Rate	Letest	0.73%	
Equity Risk Preimium	5 Years	6.74%	
Cost of Equity		7.46%	

Year	Expected Dividend&Buyback	Cumulative PV Factor (Risk-free Rate + Equity Risk Premium)	Present Value of Expected Dividends & BuyBacks
2024	¥ 2,135.77	0.931	¥1,987.47
2025	¥ 2,230.56	0.866	¥1,931.55
2026	¥ 2,329.57	0.806	¥1,877.20
2027	¥ 2,432.96	0.750	¥1,824.38
2028	¥ 2,540.95	0.698	¥1,773.05
2029	¥ 2,653.73	0.649	¥1,723.16
2030	¥ 2,771.51	0.604	¥1,674.68
2031	¥ 2,894.52	0.562	¥1,627.56
2032	¥ 3,022.99	0.523	¥1,581.77
2032 - ∞	¥ 46,870.04	0.523	¥ 24,524.55

5.1 Sources

- Investing.com
- Nikkei official website
- Market Premia website
- Prof. Aswath Damodaran website

5.2 Disclaimer

- The Valuations carried out are based out of good amount of assumptions which are built based on the understanding of the current scenario in Indian Capital markets.
- The report has been prepared solely for educational purpose and not emanating from the professional in practice.
- The author assumes no liability or responsibility for the losses caused by using this report as an investment advice.
- Although, every endeavor has been made to bring this report as accurate as possible, users should not rely on this report for investment purposes.

5.3 End Note

- I would like to thank the readers for their time dedicated towards reading the report. Feedbacks are always welcome. Please feel free to get in touch at jaybhingradiya18@gmail.com for any questions and/or feedback.

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I would like to extend my gratitude towards Mr. Parth Verma sir for guiding me to prepare the Index Valuation Report!